Lucas Hemming

Ames, IA | 6087727047 | lechemming@outlook.com

Education

Iowa State University | Ames, IA *Bachelor of Science in Electrical Engineering* | Expected Graduation: Summer 2026

- **Relevant Coursework:** Analog VLSI Design, Embedded Systems, Computer Architecture, Signals and Systems, Microelectronics, High-Speed Systems, Machine Learning(Machine Vision).

Technical Skills:

- **Languages:** C++, C, VHDL, Verilog, Python, MATLAB.
- **EDA tools:** Cadence Virtuoso, Open EDA tools(Magic/Xschem/ngspice), KiCad.
- **Tools:** Oscilloscopes, Logic Analyzers, Soldering.

Work Experience:

Computer Technician | ISU Surplus ***Ames, IA | August 2023 – May 2024***

- Diagnosed and repaired a variety of computer hardware and peripherals, ensuring functional equipment was ready for resale or university reuse.
- Performed secure data destruction and component-level troubleshooting on incoming hardware shipments.
- Managed inventory and provided technical support for customers and university departments regarding hardware specifications.

Team Leader | FedEx Express ***Location (Madison, WI) | 2019 – 2021***

- Led a team in a fast-paced logistics environment, overseeing daily operations to meet strict delivery and sorting deadlines.
- Coordinated workflow and resource allocation, ensuring safety protocols and operational efficiency were maintained.
- Trained new employees on company software and equipment.

Key Projects:

Sky130nm Open Source Design & Documentation

- Worked in a team to design and implement various basic integrated components(DAC, OP-AMP)
- Utilized open-source EDA tools (**Magic, Netgen, Xschem**) for schematic/layout extraction simulation loop.
- Optimized layout and managed resources for April 2026 Tapeout.
- Wrote extensive documentation for ASIC curious university students unfamiliar with open-source tools.
- Designed testing suite for post fabrication characterization and comparison to simulation.

Sub-Nanosecond TDR Pulse Generator

- Designed and laid out a 4-layer PCB in KiCad re-purposing commercial IC’s to generate sub-200ps edges for Time Domain Reflectometry.
- Performed detailed signal integrity analysis including controlled-impedance trace routing, differential pair matching, and decoupling strategy informed by manufacturer datasheets and reference designs.
- Performed post-fabrication testing to characterize performance of design and make adjustments.